Mathematical Structures and Tables

Technical Writing Practical

Grouped Expression

$$\overbrace{a+b+c}^{\text{First Group}}+\overbrace{x+y+z}^{\text{Second Group}}=\overbrace{a+\underbrace{b+c}_{\text{Sum 1}}}^{\text{Total}}$$

Array

$$A=\left[\begin{array}{c|ccc} & x_1 & x_2 & x_3 \\ \hline y_1 & a_{11} & a_{12} & a_{13} \\ y_2 & a_{21} & a_{22} & a_{23} \\ y_3 & a_{31} & a_{32} & a_{33} \end{array}\right]$$

Piecewise Function

$$f(x)=\begin{cases} x^2+2x+1, & x<0, \\ \sqrt{x+1}, & 0\le x<4, \\ x-1, & x\ge 4. \end{cases}$$

Matrix with Column Groups

$$\begin{bmatrix} \overbrace{a_{11} & a_{12} & a_{13}}^{\text{Column Group 1}} & \overbrace{a_{14} & a_{15}}^{\text{Column Group 2}} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \end{bmatrix}$$

Quadratic and Matrix Contributions

$$F(x)=\underbrace{\sum_{i=1}^{n}(a_i x_i^2+b_i x_i+c_i)}_{\text{Quadratic Contribution}}+\underbrace{\begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} A \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{\text{Matrix Contribution}}$$

$$A=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

Student Table

Roll No.	Name	Department	Semester	CGPA
101	Amit	CSE	2	8.45
102	Neha	CSE	2	9.12
103	Rahul	ECE	4	8.76
104	Priya	ECE	4	9.01

Multicolumn and Multirow Tables

Academic Performance		
Student	Midterm	Final
A	78	89
B	85	92
C	91	94

Department	Course	Credits	Grade
CSE	Data Structures	4	A
	Machine Learning	4	A+
	Operating Systems	3	B+
ECE	Digital Electronics	4	A
	Signals	3	B+

Scientific Table

Model	Parameters	FLOPs	Accuracy	Energy (kWh)
ResNet18	11.7M	1.8G	94.2%	0.00093
MobileNetV2	3.5M	0.3G	92.8%	0.00061
EfficientNet-B0	5.3M	0.39G	93.5%	0.00072

Student Performance with Conditional Colours

Student	Mathematics	Physics	Computer Science
Amit	92	81	95
Neha	72	88	91
Rahul	68	74	79
Priya	96	94	98

Paragraphs in Cells

Technique	Description	Advantages	Limitations
Pruning	Removes unnecessary parameters from a neural network.	Reduces model size and computation.	May reduce accuracy.
Quantization	Represents weights using lower precision.	Reduces memory and energy consumption.	Can introduce numerical error.
Knowledge Distillation	Transfers knowledge from a large teacher model to a smaller student model.	Produces compact models.	Requires training a teacher model.

Professional Table

Table 1: Comparison of model performance

Model	Parameters	FLOPs	Accuracy	Energy
ResNet18	11.7M	1.82G	94.21	0.00093
MobileNetV2	3.5M	0.30G	92.84	0.00061
EfficientNet-B0	5.3M	0.39G	93.52	0.00072

Nested Table

Model	Architecture	Performance
CNN	Conv → ReLU → Pool	Metric Value Accuracy 94% F1 Score 93%
ResNet	Conv → BN → ReLU	Metric Value Accuracy 96% F1 Score 95%
ViT	Attention → MLP	Metric Value Accuracy 97% F1 Score 96%

Mathematical Expressions in a Table

Model	Loss Function	Parameters	Complexity
Linear	$\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$	10K	$O(n)$
CNN	$-\sum_i y_i \log(\hat{y}_i)$	1.2M	$O(n^2)$
Transformer	$-\sum_i y_i \log p_i$	110M	$O(n^2 d)$